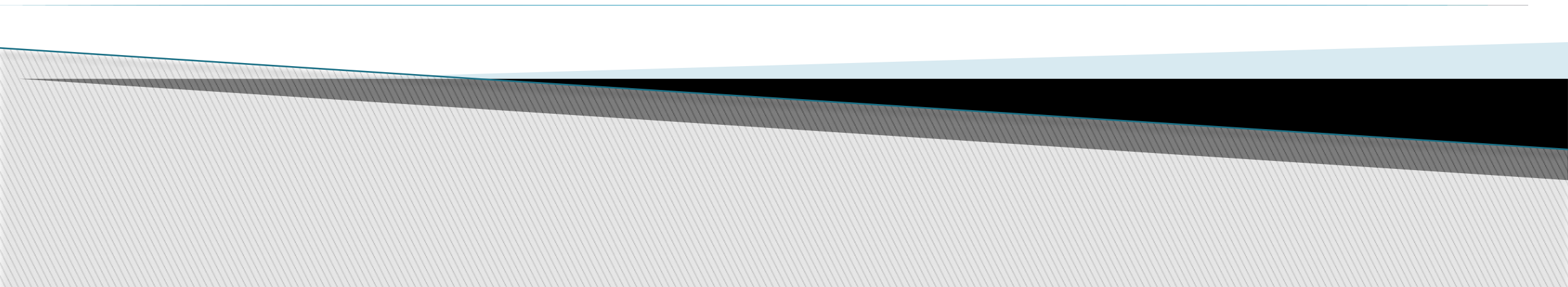


Hemodynamic Disorders Thromboembolic Disease

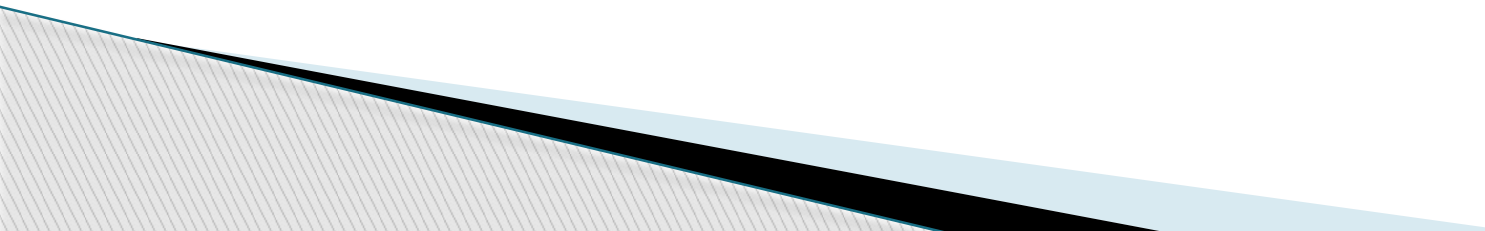
Dr. Bethel Fekede

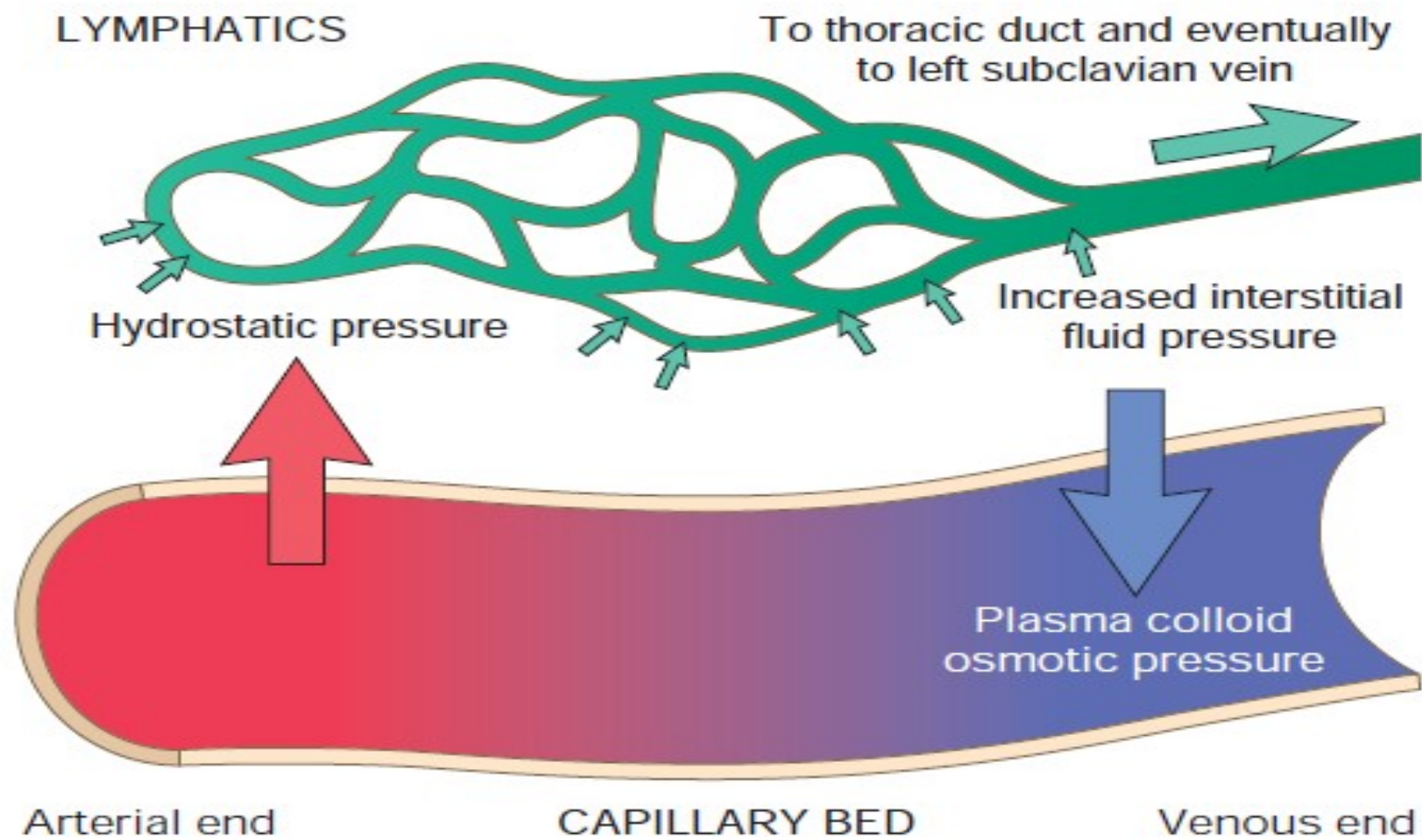


Hemostasis

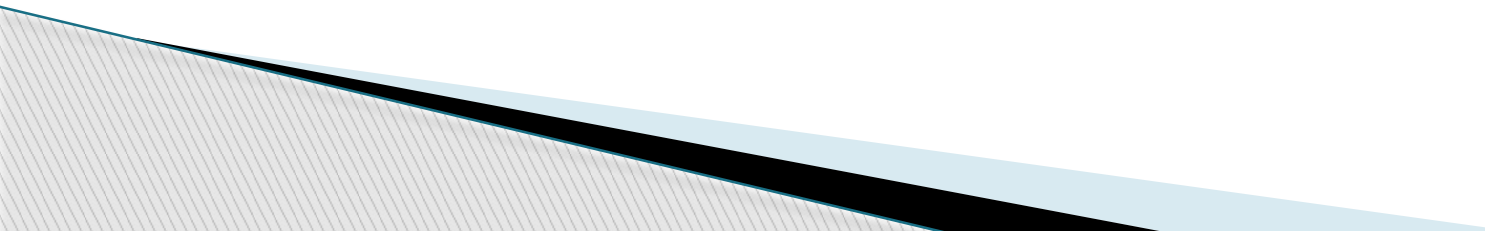
- ▶ Platelets
- ▶ Clotting factors
- ▶ fibrinolytics

Hemodynamic Disorders

- ▶ Edema----fluid in tissue
 - ▶ Effusion----fluid in body cavity
 - ▶ Shock-----a state of systemic hypoperfusion
 - ▶ Hemorrhage----bleeding(internal/external)
- 



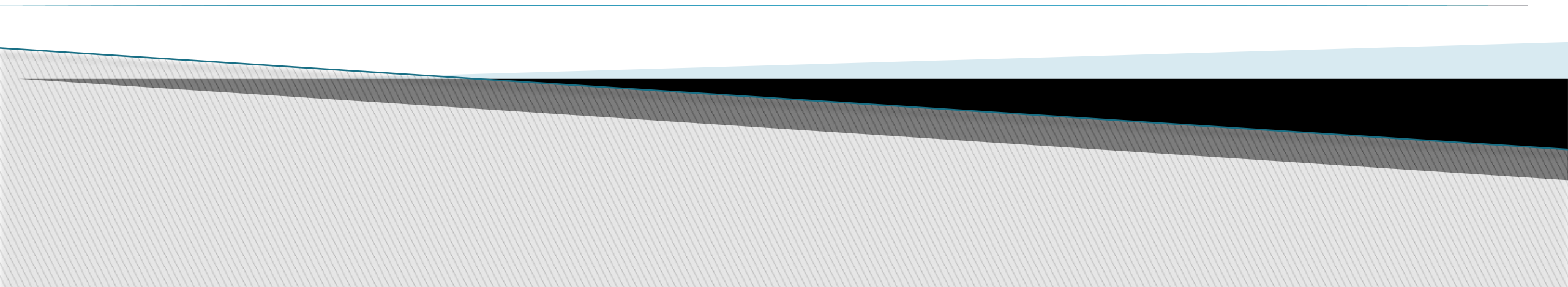
Edema: could develop due to

- ▶ *Increased Hydrostatic Pressure*
 - ▶ *Reduced Plasma Osmotic Pressure*
 - ▶ *Sodium and Water Retention*
 - ▶ *Lymphatic Obstruction*
- 

Shock

“Cardiovascular collapse”

Systemic hypoperfusion inability to adequately perfuse the whole body,
for any reason



Shock

- ▶ Caused by **reduction** either in **cardiac output** or in **effective circulating blood volume**
- ▶ Hypotension, impaired tissue perfusion & cellular hypoxia
- ▶ Clinically manifested by
 - Hemodynamic disturbances
 - Organ dysfunction

Causes of shock

1. **Cardiogenic shock** (i.e., pump failure)

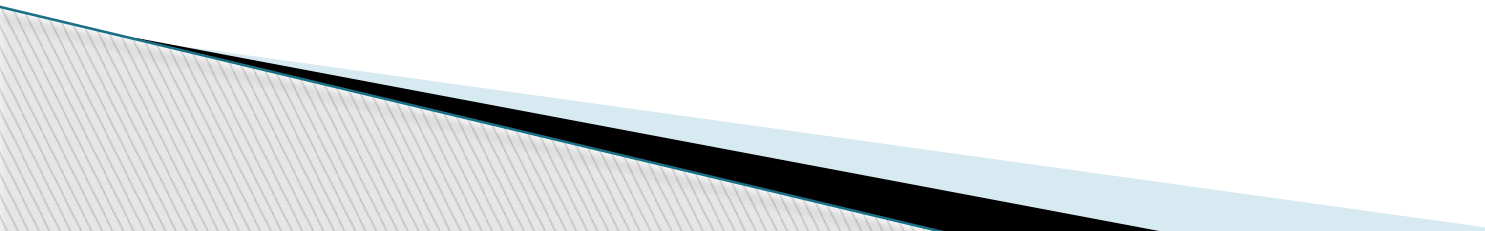
Examples: massive myocardial infarct, rupture (ventricle, valve), diphtheria, arrhythmias

2. LOSS OF VASCULAR TONE (Distributive)

- I. **Septic shock** (i.e., from bacterial breakdown products and especially cytokine production by way of nitric oxides)
- II. **Anaphylaxis**: IgE mediated hypersensitivity response (generalized mast-cell degranulation)
- III. **Neurogenic**: certain poisons (notably war gases), Profound anaesthesia, Spinal cord injury, Vasovagal (i.e., extreme pain, emotion)

Causes of shock

3. Hypovolemic shock

- ▶ Bleeding: Externally or Internally (GI bleeds, hemoperitoneum)
 - ▶ Other fluid loss: sweating, vomiting, diarrhea, Burns
 - ▶ Third-space losses (i.e., into effusions or ileus)
- 

Causes of shock

4. Obstructive Shock

=> Pulmonary embolism, aortic aneurysm, cardiac tamponade

Effects shock

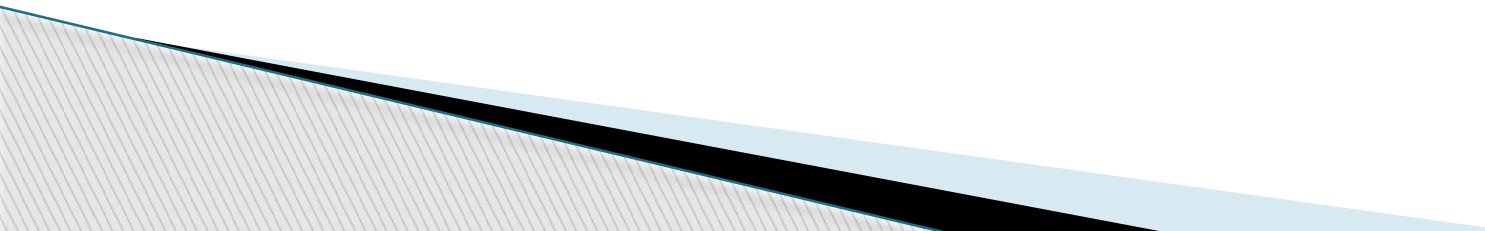
- ▶ Regardless of etiology, hypoperfusion of the whole body produces common problems
- ▶ A variety of other secondary mediators of shock are produced by hypoperfused tissues to further exacerbate the problem
- ▶ Histamine, serotonin, leukotrienes, TNF, IL1, C3a, C5a, => dilate vessels, inviting blood to pool in venules and/or make small vessels permeable, causing fluid to leak out.

With widespread anaerobiosis, lactic acidosis is likely to develop, and pH goes way down

When endothelial cells are damaged or thromboplastin enters the circulation, DIC may result

Stages of shock

- ▶ Shock is a **progressive disorder** if uncorrected leads to death
- ▶ Three phases
 1. Non progressive phase (compensated shock):
 - => Reflex **compensatory mechanism activated** & perfusion of **vital organs** maintained
 - ✓ Baroreceptor reflexes, release of catecholamines, activation of renin angiotensin axis, antidiuretic hormone release, & generalized sympathetic stimulation

- ✓ Blood is **shunted** away from the kidneys, salivary glands, gut, skin, and muscles in order to perfuse the **brain and heart**.
 - ✓ Blood pressure is maintained
 - ✓ Patients may be **oliguric** from reduced blood flow to the kidneys ("prerenal azotemia"), and have **dry mouth and skin** from reduced blood flow to these organs
- 

Stages of shock

2. Progressive stage (decompensated shock):

= Tissue hypo perfusion

= Worsening circulatory & metabolic imbalances :
blood pressure and cardiac output decline

- ▶ Lactic acidosis lowers the tissue pH & arteriolar dilatation => blood pools in microcirculation => decrease CO & endothelial injury => DIC

Stages of shock

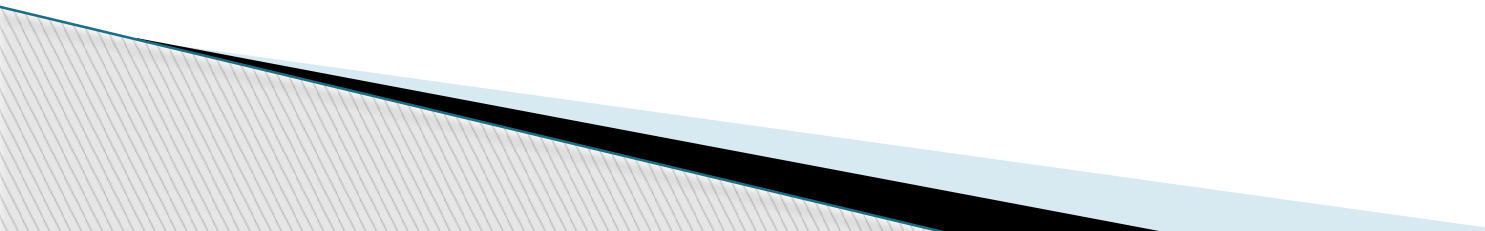
- 3. **Irreversible stage**: severe cellular injury (lysosomal enzyme leakage) & myocardial contractile dysfunction worsens & ischemic bowel allows bacteria to enter circulation
- ✓ Correction of the deficient cardiac output and volume deficit by any means **fail to reverse shock**
- ✓ Blood pressure and pH continue to drop
- ✓ **Death due to multiple organ failure (MOF)**

Septic shock

- ▶ **Systemic microbial infection** e.g gram negative infections (endotoxic shock)
- ▶ 25–50% mortality
- ▶ Results from **spread of localized infection** e.g abscess, peritonitis, pneumonia into blood stream
- ▶ 70% are caused by endotoxin producing gram negative bacteria **“endotoxic shock”**
- ▶ Endotoxins are bacterial wall **lipopolysaccharides (LPSs)**

Morphology

- ▶ Tissue injury is **hypoxic injury**
- ▶ Any tissue
- ▶ Diffuse hypoxic injury to the **brain** ("respirator brain")
- ▶ **Subendocardial** necrosis
- ▶ Widespread contraction **band necrosis** in the heart
- ▶ **Acute tubular necrosis** of the kidneys

- ▶ Stressed adrenals, i.e., cortical hyperplasia, lipid depletion
 - ▶ Bleeding points ("stress ulcers") in the stomach and duodenum
 - ▶ Necrosis of, and bleeding into, portions of the small intestine (especially when dopamine has been administered)
 - ▶ fatty change of the liver and centri-lobular necrosis
- 

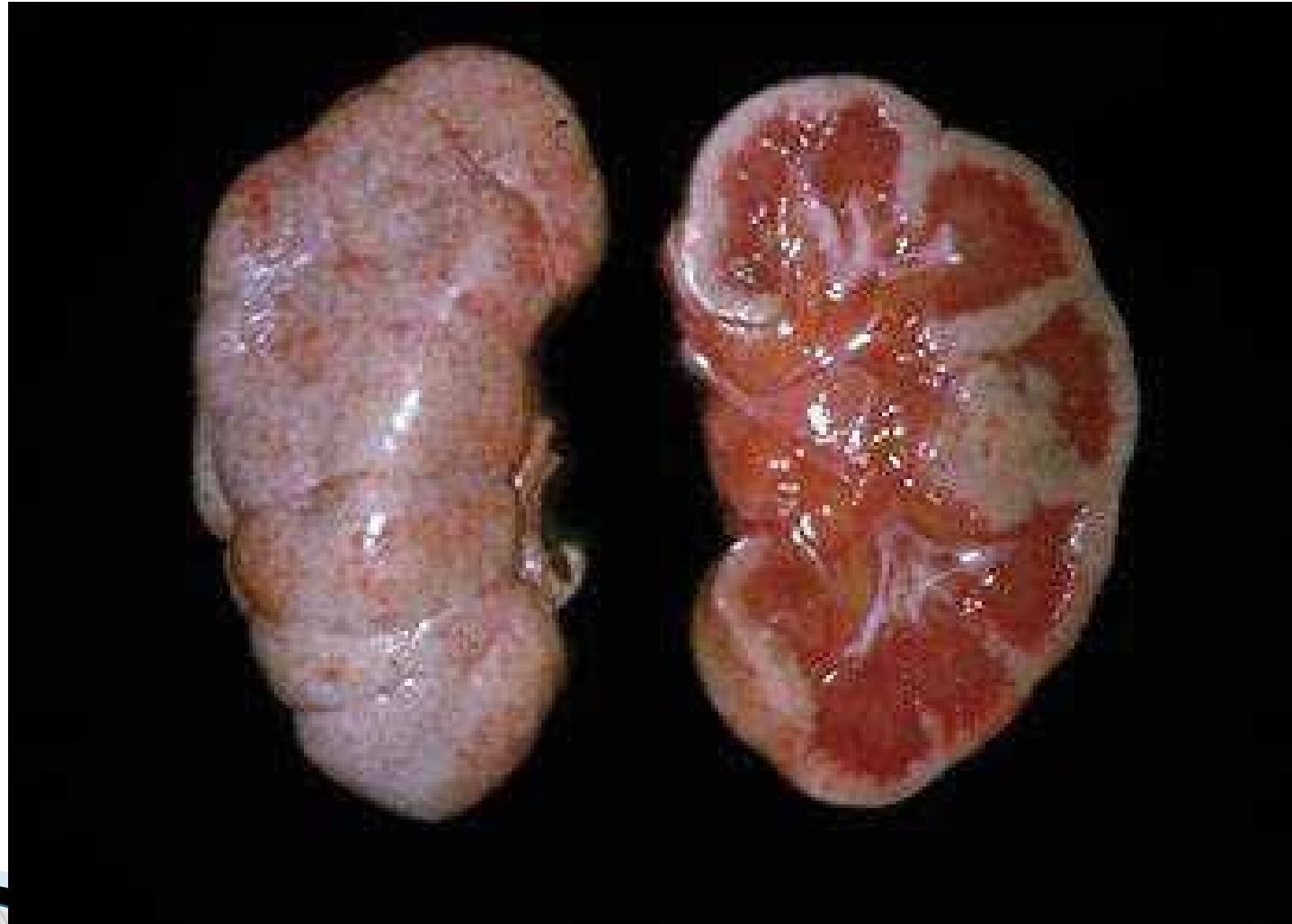
Watershed Infarcts

... after shock (systemic hypoperfusion)

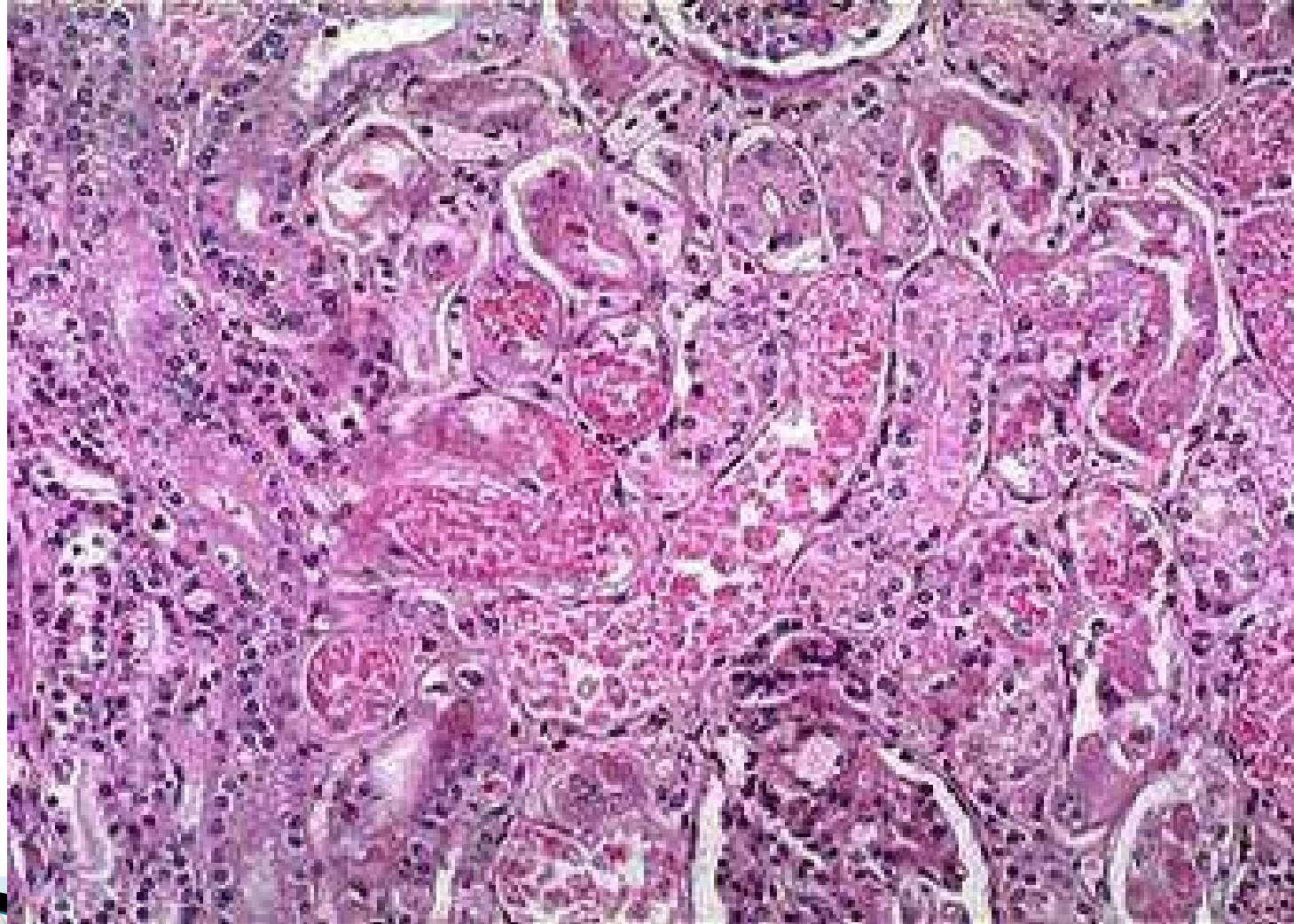


Necrosis at the places which are farthest from the arterial supply.

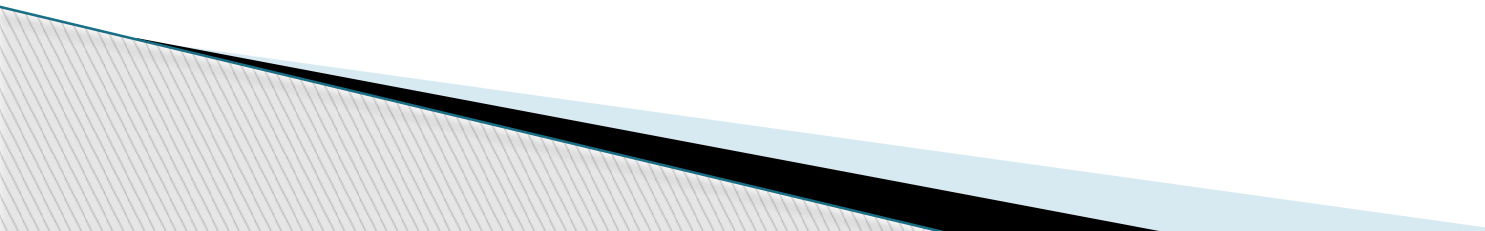
Kidney, renal tubular necrosis due to shock



Kidney, acute tubular necrosis due to shock



HEMORRHAGE

- ▶ **Hemorrhage**: Blood cells that have escaped from a vessel
 - ▶ Trauma, atherosclerosis, inflammatory or neoplastic erosion, chronic congestion...
 - ▶ **Hemorrhagic diathesis** : insignificant injury
 - ▶ **Hematomas**: bleeding enclosed in tissues
- 

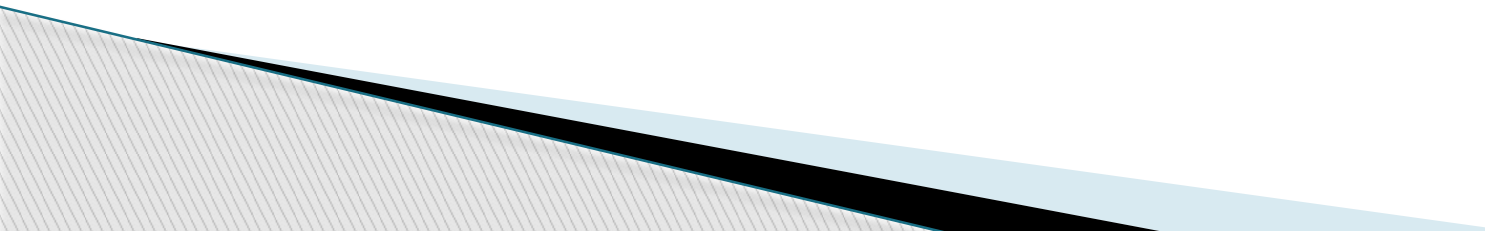
- ▶ Hemorrhage is grouped according to size
- ▶ Large amount of bleeding in to:

Hemothorax: Blood in a pleural cavity.

Hemopericardium: Blood in the pericardial cavity

Hemoperitoneum: Blood in the peritoneal cavity

Hemarthrosis: Bleeding into a joint



Petechie : <3mm , mucous membrane, skin, serosal surfaces.

Increased intravascular pressure, thrombocytopenia, clotting factor deficits, defective platelet function

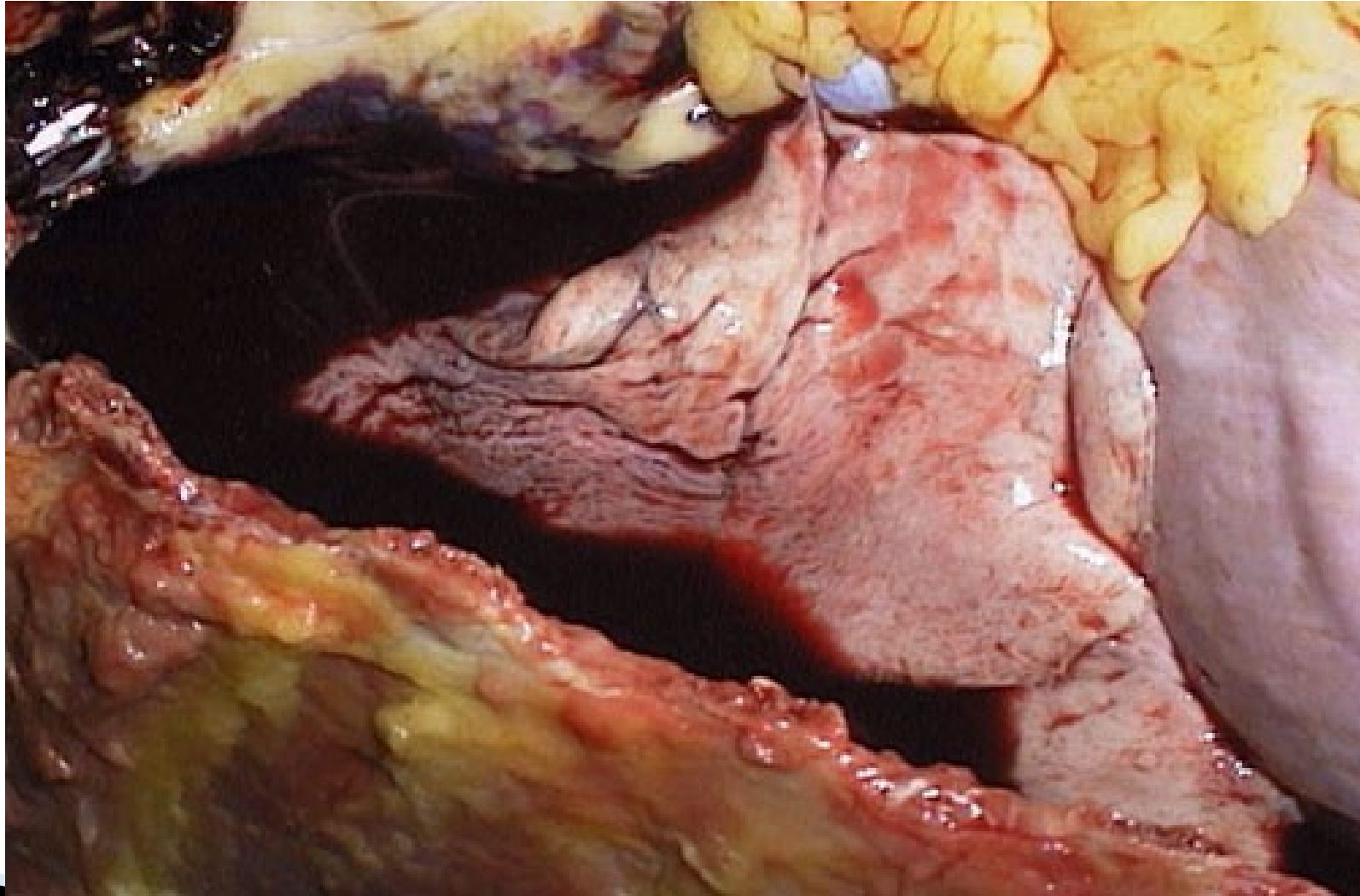
Purpura: $\geq 3\text{mm}$, trauma, local vascular inflammation, increased vascular fragility + above causes

Ecchymosis: $> 1\text{cm}$, subcutaneous hematomas, AKA bruise, trauma typical cause + above causes

Hematomas



hemothorax



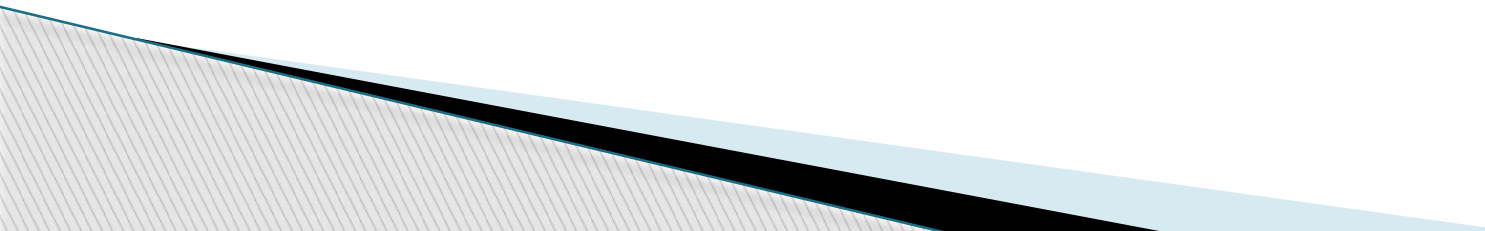
Ecchymoses:



Petechiae



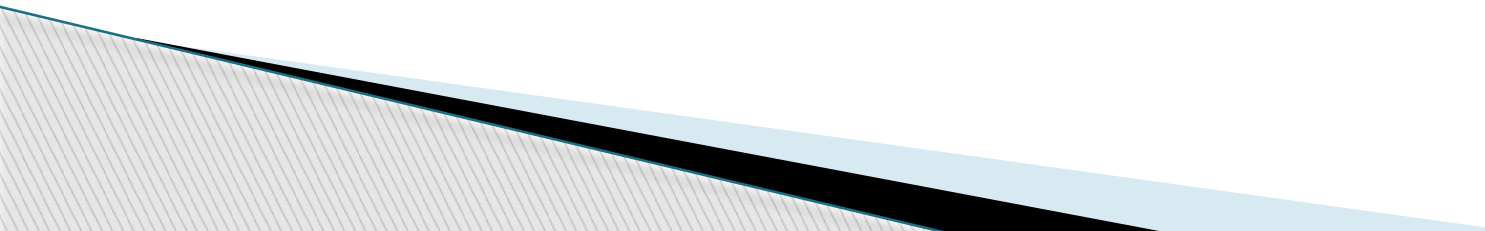
Bleeding disorders

1. Increased fragility of vessels
 2. Platelet deficiency (number) or dysfunction (adhesion, aggregation & secretion)
 3. Derangement of coagulation
 4. Combination of these
- 

Increased fragility of vessels (Nonthrombocytopenic purpura)

▶ Petechia & purpura

Causes:

1. Infections e.g., meningococcemia
 2. Drug reactions
 3. Scurvy
 4. Amyloidosis
 5. Henoch–Schonlein purpura
 6. Hereditary hemorrhagic telangiectasia
- 

Thrombocytopenia

- ▶ $<100,00/\mu\text{l}$
- ▶ Prolonged bleeding time & normal PT & PTT
- ▶ Hemorrhage in Skin, GIT & GUT
- ▶ Causes
 1. Decreased production: aplastic anemia & leukaemias, etc
 2. Decreased survival: immunologic autoimmune (idiopathic, drug, HIV) or nonimmunologic (mechanical)
 3. Sequestration: hypersplenism
 4. Dilutional

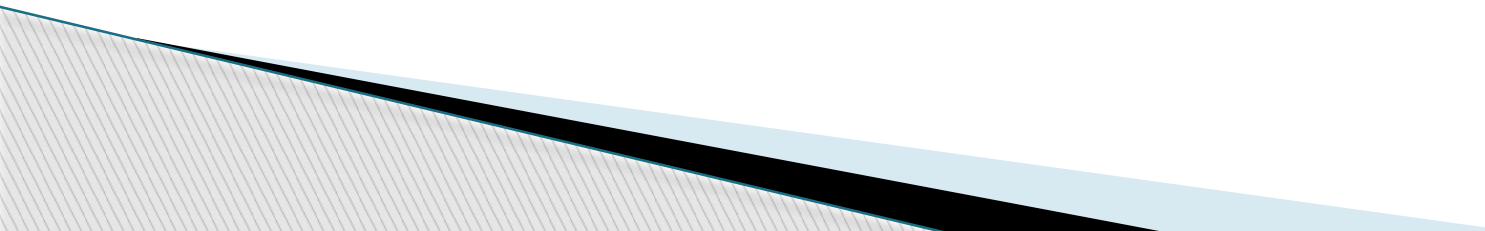
Abnormality in clotting factors

- ▶ Ecchymoses or prolonged bleeding after trauma, petechiae are absent
- ▶ Hereditary (hemophilia, Von Willebrand disease) or acquired (Vit. K deficiency, Liver disease)

Hemophilia



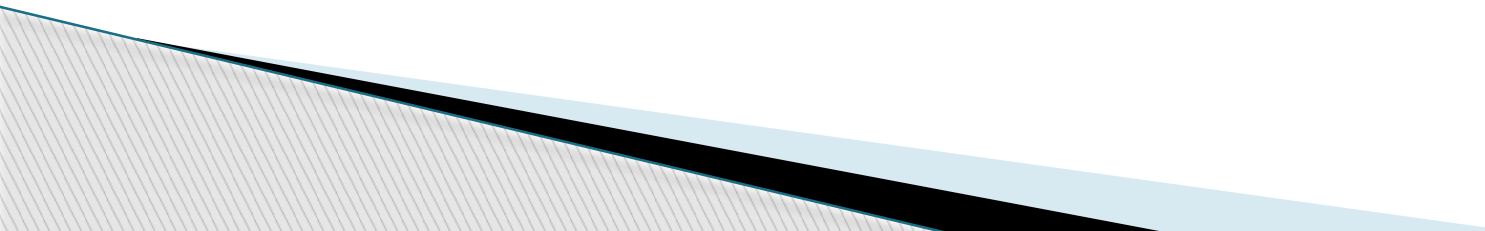
Disseminated intravascular coagulation (DIC)

- ▶ “Consumption coagulopathy”
 - ▶ is acute, subacute or chronic thrombo-hemorrhagic disorder in variety of diseases(secondary complication)
 - ▶ Microthrombi in microcirculation
 - ▶ Consumption of platelets, fibrin, coagulation factors & activation of fibrinolytic mechanisms=> hemorrhage
- 

Presentation

- ▶ SSx of infarction=> microthrombi
- ▶ Hemorrhagic tendencies=> activation of lytic pathway + depletion of clot forming substances.

Mechanisms of DIC

1. Release of tissue factor or thromboplastic substance
 2. Wide spread injury to endothelium
 - Obstetrics complications (abruptio placentae, toxemia, etc),
 - infections (gram negative sepsis, etc),
 - Neoplasms
 - massive tissue injury (burn, trauma)
- 

Thank You

